10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 19 — Electronics

Solved Exercise | New Book (PECTAA 2026)

Exercise 19 (Solved)

This exercise covers the semiconductor diode, forward and reverse bias, the light-emitting diode, analog and digital electronics, Boolean logic, logic gates and their use in burglar and fire alarms.

A. Multiple Choice Questions

19.1 A diode allows current to flow in:

(a) both directions
(b) neither direction
(c) one direction only
(d) random directions

Answer: (c) one direction only — a diode conducts when it is forward biased and blocks the current when it is reverse biased, so it acts as a one-way valve for current.

19.2 What is the primary function of a light-emitting diode?

(a) To block current in both directions
(b) To emit light when current passes in one direction
(c) To amplify electrical signals
(d) To convert light into electrical energy

Answer: (b) To emit light when current passes in one direction — in an LED, electrons and holes recombine at the junction in forward bias and the energy is given out as light; it does not conduct in reverse bias.

19.3 The output of a two inputs NOR gate is '1', when:

(a) A is '1' and B is '0'
(b) A is '0' and B is '1'
(c) both A and B are '0'
(d) both A and B are '1'

Answer: (c) both A and B are '0' — a NOR gate is an OR gate followed by a NOT gate, so its output is 1 only when the OR output is 0 — that is, when both inputs are 0.

19.4 The output of a NAND gate is '0', when:

(a) both of its inputs are '0'
(b) both of its inputs are '1'
(c) any of its inputs is '0'
(d) any of its inputs is '1'

Answer: (b) both of its inputs are '1' — NAND = NOT + AND; the AND output is 1 only when both inputs are 1, and the NOT inverts it, so the output is 0 exactly in that case.

19.5 How many bits make up a byte in Boolean logic?

(a) 2
(b) 4
(c) 6
(d) 8

Answer: (d) 8 — a bit is a single binary digit (0 or 1) and eight bits together make one byte.

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B. Short Answer Questions

19.1 What is the function of a semiconductor diode?

A semiconductor diode is a two-terminal device made of a p-n junction. Its function is to allow current to flow through it in one direction only, so it works as a one-way valve (rectifier).

When the p-side is connected to the positive terminal (forward bias), the diode conducts easily and the current flows. When the connections are reversed (reverse bias), the diode blocks the current almost completely. It is therefore used to change alternating current into direct current in a rectifier, to protect circuits from being connected the wrong way round, and to detect radio signals.

19.2 What happens when a diode is forward biased?

When a diode is forward biased, its p-type region is connected to the positive terminal of the battery and its n-type region to the negative terminal. The applied field pushes the holes in the p-region towards the junction and the electrons in the n-region towards the junction too.

As a result the depletion region (the barrier layer) becomes thinner and the potential barrier is lowered. When the applied voltage is greater than about 0.7 V for silicon (0.3 V for germanium), the barrier is overcome and a large current flows through the junction. The diode then behaves almost like a closed switch with a very small resistance, and the current increases rapidly with a further rise in voltage.

19.3 What is the function of an LED?

An LED (light-emitting diode) converts electrical energy into light energy. It is a specially made p-n junction of a material such as gallium arsenide phosphide, and when a current passes through it in forward bias it emits light.

Its functions and uses: it acts as an indicator lamp in TVs, radios, mobile chargers, switches and instrument panels; it is used in digital displays and seven-segment numbers, in traffic signals, in torches and in large display screens because it is bright, small, rugged and needs very little power. It works only in forward bias, so it also indicates the direction of current in a circuit.

19.4 Why does an LED emit light when current passes through it?

In an LED, when the junction is forward biased, electrons from the n-region and holes from the p-region are pushed towards the junction and cross it in large numbers. Near the junction an electron falls into a hole and recombines with it. The electron moves from a higher energy level (conduction band) to a lower one (valence band), and the difference of energy is given out.

In an ordinary silicon diode this energy appears as heat, but in an LED the semiconductor is chosen from a material with a suitable energy gap — a direct band gap — so that the energy is emitted as a photon of visible light. The colour of the light depends on the size of the energy gap: a larger gap gives blue light, a smaller gap red or infrared. That is why an LED lights up when current flows through it and why it does not conduct in reverse bias.

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19.5 What does an analog-to-digital converter (ADC) do?

An analog-to-digital converter changes a continuously varying (analog) signal — such as a sound wave from a microphone, a voltage from a sensor, or an electrical signal from a thermocouple — into a stream of binary numbers (0s and 1s) that a computer or digital circuit can process.

It works in stages: it samples the analog signal at regular intervals, holds each value, and then quantises it — comparing it with a set of reference levels and rounding it to the nearest one — before coding it into a binary word of a fixed number of bits. The process is called sampling, quantisation and encoding.

ADCs are needed because most signals in the real world are analog while computers and digital instruments work with binary numbers. They are used in digital multimeters, sound recording equipment, mobile phones, digital cameras and in every sensor that is connected to a microcontroller.

19.6 What is the difference between analog and digital signals?

Point	Analog signal	Digital signal
Nature	continuous; can take any value within a range	discrete; takes only two levels, 0 (low) or 1 (high)
Representation	a smooth, wave-like curve	a square wave / a series of pulses or binary numbers
Examples	sound from a microphone, the output of an LDR, a thermometer reading, the old record player	output of a computer, CD and DVD data, output of logic gates, digital TV signals
Noise	noise is added to the signal and cannot be removed	noise and small changes can be corrected, because only 0 and 1 matter
Storage and processing	difficult to store and copy without loss	easy to store, copy and process without loss of quality
Transmission over long distances	weakens and gets distorted; needs amplification which also amplifies noise	very reliable — the pulses can be regenerated exactly

In short, an analog signal carries information in the exact size of a continuous quantity, while a digital signal carries it as a coded series of 0s and 1s. Analog-to-digital converters are used to change one into the other.

19.7 What is Boolean logic in digital electronics?

Boolean logic is a system of logic in which every statement or signal can have only one of two values: true or false, which in electronics are written as 1 and 0 (high and low voltage). It was developed by the mathematician George Boole and uses the operators AND, OR and NOT, from which NAND and NOR are formed.

In digital electronics Boolean logic is the basis of all decision-making circuits. Logic gates are the physical devices that perform these operations, and combining them lets a circuit decide, for example, to sound an alarm only when two conditions happen at the same time, or when either one of them happens. Since every piece of data in a computer, phone or digital instrument is a pattern of 0s and 1s, the whole of digital technology — computers, calculators, washing machine controllers and traffic lights — is built upon Boolean logic.

19.8 What does a NOT gate do in a digital circuit?

A NOT gate has one input and one output, and it simply inverts the input: if the input is 1 its output is 0, and if the input is 0 its output is 1. That is why it is also called an inverter. Its Boolean expression is $Y = \bar{A}$ (A bar).

Its uses: it turns a signal into its opposite, so a switch that gives 1 when pressed can be used to give 0; two NOT gates in series restore the original signal as a buffer; and it is the gate that turns an AND into a NAND and an OR into a NOR. In burglar alarms and safety circuits the NOT gate is what makes the alarm sound when the sensor goes low (for example when the light beam is cut).

19.9 What is the role of an AND gate in digital circuits?

An AND gate has two or more inputs and one output. Its output is 1 only when all of its inputs are 1; if any input is 0, the output is 0. Its Boolean expression is $Y = A \cdot B$.

Its role is to act as a coincidence detector — it gives a signal only when two conditions are satisfied at the same time. Examples: in a washing machine the motor starts only when the door is closed AND the start button is pressed; in a car the ignition works only when the key is in AND the gear is in neutral; in a security lock two switches must be pressed together; and in a fire alarm system an AND gate confirms a fire when both the smoke sensor AND the heat sensor respond. Together with OR and NOT gates it can be combined to build any decision-making circuit.

C. Constructed Response Questions

19.1 Why would a diode be damaged if it is connected in reverse bias with a very high voltage? What does this tell us about the diode's physical limits?

In reverse bias the depletion region of the diode becomes wider and the small current that flows (the leakage current) is due only to the few minority carriers. The diode normally blocks the current, and the junction is designed to withstand only a certain reverse voltage, called the peak inverse voltage (PIV).

If the reverse voltage is increased too much, the electric field across the narrow junction becomes so strong that it tears electrons out of their atoms — avalanche breakdown — or pulls electrons directly across the barrier (Zener breakdown). The current then rises suddenly to a very large value, and the power I^2R dissipated at the junction makes it extremely hot. The junction melts or the crystal is destroyed, so the diode is permanently damaged and may become a short circuit.

This shows that a diode has physical limits set by the material and the thickness of the depletion layer: a maximum reverse voltage, a maximum forward current and a

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maximum power it can dissipate. In practice a diode is protected by connecting a resistor in series to limit the current, and by choosing a diode whose PIV and current rating exceed the circuit values; a Zener diode is deliberately operated in the controlled breakdown region to give a steady reference voltage.

19.2 What makes an LED glow brighter or dimmer depending on the voltage applied? How is this property used in adjustable display brightness?

The brightness of an LED depends on the number of electrons and holes that recombine at the junction per second, and this depends on the current flowing through it. In the forward direction the diode current rises very rapidly with the applied voltage (an exponential relation), so a small increase of voltage causes a large increase of current and the LED glows much more brightly. Reducing the voltage decreases the current and the LED becomes dimmer; below the threshold voltage (about 1.8–3 V depending on the colour) it does not light at all.

Because the brightness follows the current, the LED is always used with a series resistor. For adjustable brightness the current is controlled in one of two ways: a variable resistor (potentiometer or rheostat) in series with the LED, so that turning the knob changes the resistance and hence the current; or a pulse-width modulation (PWM) circuit, which switches the LED on and off thousands of times a second and changes the fraction of time it is on — the eye sees this as a steady but dimmer light, and the setting can be made with a digital control.

The same method is used for the brightness control of mobile phone screens, laptop displays, TV backlights and car dashboards: the LEDs are driven with a controlled current or a controlled duty cycle, so the display brightness can be changed smoothly without any change in the colour of the light.

19.3 Why are digital systems more reliable than analog systems when transmitting signals over long distances?

In an analog system the information is carried by the exact size of a continuously varying voltage. Over a long cable or a radio link the signal becomes weaker and picks up noise and interference; when it is amplified to make up for the loss, the noise is amplified along with it. Because the correct value is not known at the receiver, the noise cannot be removed, and the quality keeps falling with distance — this is heard as hiss in an old analog broadcast.

In a digital system the information is carried only as two levels, 0 (low) and 1 (high). At the receiver a threshold level decides whether a pulse is a 0 or a 1. Even if noise changes the height of the pulse somewhat, as long as it stays on the correct side of the threshold the receiver recovers exactly the original signal; the pulse is then regenerated at full strength by a repeater and sent on. So no noise accumulates, however far the signal travels.

Digital systems can also detect and correct errors by adding check bits, and they can compress, store and copy the data without loss of quality. This is why a CD, a digital telephone call or a satellite TV broadcast is clear even after a long journey, while the older analog versions would have become noisy. The only requirement is

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enough bandwidth to send the extra pulses.

19.4 Why is the NOT gate also called an inverter, and how can such a simple function have a major role in digital decision-making?

The NOT gate is called an inverter because its only job is to invert the signal: input 1 gives output 0 and input 0 gives output 1. The output is always the opposite of the input, that is why the name inverter is used.

Its importance comes from the fact that decisions often have to be reversed. A sensor that gives 1 when it is safe must be turned into a 0 to sound an alarm — the NOT gate does this. Again, some signals are active-low: the switch gives 0 when pressed, and the circuit needs 1, so a NOT gate is used to match them.

Because every logic family is built from AND, OR and NOT, and the last of the three is the cheapest and simplest, it appears everywhere: it makes NAND from AND and NOR from OR; it creates the alternating clock signals of a computer; it can be used as a buffer to strengthen a weak signal; and it is the gate that decides 'if not A then do B' in burglar alarms, safety interlocks, traffic-light controllers and computer logic. A single inverter can therefore change the whole sense of a decision, which is why it has a major role in digital decision-making.

19.5 Why would a security system use NAND gates instead of OR or AND gates for certain logical conditions?

A NAND gate gives a 0 output only when all its inputs are 1; in every other case its output is 1. This is exactly the behaviour needed in many security and alarm systems, where the machinery or condition is 'alarm sounds when the system is NOT satisfied'.

A burglar alarm is a good example. Light falling on the LDR makes its resistance low, so input B is high (1); with the push-button switch S not pressed, input A is high (1) too. With both inputs high, the NAND output is low and the alarm is off. If the burglar blocks the light (B becomes 0), or steps on the switch (A becomes 0), or if both happen, the output goes high and the alarm rings. So with a NAND gate the alarm is fired by the failure of a condition, which is the natural, fail-safe behaviour for security — and the same gate can be built cheaply from simple transistors.

Further, NAND is called a universal gate, because AND, OR and NOT can all be made from NAND gates alone; a manufacturer can therefore build a whole alarm system from a single type of chip, which reduces cost and simplifies the circuit. An OR gate would sound the alarm even in the safe condition and an AND gate would need both inputs to be high to act, so neither matches the logic of 'alarm when the guard condition fails' as directly as the NAND gate does.

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D. Comprehensive Questions

19.1 Explain how the depletion region of a diode changes in forward and reverse bias. What effect does this have on current flow?

When a p-n junction is formed, electrons from the n-side diffuse into the p-side and holes from the p-side into the n-side, and they recombine near the junction. This leaves a layer on each side that has no free carriers but fixed charged ions — negative ions on the p-side and positive ions on the n-side. This layer is the depletion region, and the fixed charge in it sets up a potential barrier of about 0.7 V for silicon that opposes any further movement of charge.

Forward bias. The p-side is joined to the positive terminal and the n-side to the negative terminal. The applied field pushes holes and electrons towards the junction, so many carriers reach it and recombine. The depletion region becomes narrower and the barrier is lowered. When the applied voltage exceeds about 0.7 V, the barrier is overcome and a large current flows; the diode behaves like a closed switch of low resistance. The current increases rapidly (almost exponentially) with further increase of voltage.

Reverse bias. The p-side is joined to the negative terminal and the n-side to the positive terminal. The applied field pulls the holes and electrons away from the junction, so the depletion region becomes wider and the barrier becomes higher. Almost no majority carriers can cross, so only a tiny leakage current flows, made up of the few minority carriers; the diode behaves like an open switch of very high resistance. If the reverse voltage is made too large, breakdown occurs and a heavy current suddenly flows.

Effect on current flow: the diode conducts in one direction only. That is why a diode is used as a rectifier to convert a.c. into d.c., as a protective device so that a circuit is not damaged by a wrongly connected supply, and as a detector in radio receivers.

19.2 Describe how an LED produces light. Also explain how the type of semiconductor material affects the colour of the light.

Construction and working. An LED is a p-n junction made from a direct-band-gap semiconductor such as gallium arsenide (GaAs), gallium phosphide (GaP), gallium arsenide phosphide (GaAsP) or gallium nitride (GaN). It is mounted in a small plastic lens or dome that focuses the light and protects the chip, and its two leads are the anode (p-side) and the cathode (n-side).

In forward bias, electrons are injected into the p-region and holes into the n-region, and they recombine near the junction. When an electron falls from the conduction band to a hole in the valence band, the electron loses an amount of energy equal to the band gap E_g. In a direct-gap material this energy appears almost entirely as a single photon of light of frequency f such that $E_g = hf$, so the LED emits light instead of heat. The light escapes through the transparent plastic dome.

Effect of the material. The colour of the emitted light is decided by the size of the band gap, and the band gap is a property of the semiconductor material chosen. A larger band gap gives a photon of higher energy and shorter wavelength: GaN with a wide gap gives blue or even ultraviolet light; GaP and GaAsP with a moderate gap

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give green and yellow; GaAs with a small gap gives red or infrared light (used in remote controls and in optical fibres). So by choosing a different semiconductor, the manufacturer makes LEDs of red, yellow, green, blue and white light, and the same principle gives the laser diode used in CD players.

19.3 What is Boolean logic? Discuss its importance in solving complex digital problems with examples.

Boolean logic is a branch of algebra, developed by George Boole, in which variables can have only two values: true (1) or false (0). Its basic operations are AND (·), OR (+) and NOT (bar), and every statement built from them has a definite truth value. In digital electronics the two values correspond to the two voltage levels high and low, so the logic can be realised directly with transistors.

Importance. Any digital problem — however complex — can be broken into simple yes/no decisions, written as a Boolean expression, and then built from a small set of logic gates. Boolean algebra gives rules (such as $A + A = A$, $A \cdot \bar{A} = 0$, De Morgan's theorems) for simplifying these expressions, so a designer can reduce the number of gates and hence the cost, size and power consumption of a circuit. This is how the millions of gates inside a microprocessor are designed.

Examples: (i) a burglar alarm whose condition is 'alarm if the light is blocked OR the switch is pressed' is written $Y = \text{NOT}(A \cdot B)$; (ii) a washing machine that starts only if the door is closed AND the start button is pressed is $Y = A \cdot B$; (iii) a majority-vote circuit for two of three sensors is $Y = AB + BC + CA$; (iv) a lift door that must not close if someone is still in the doorway uses a NOT operation on the sensor signal. Traffic-light controllers, calculators, computers and mobile phones all work on exactly the same principle.

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19.4 What are logic gates? Describe the working of NAND and OR gates with the help of truth tables and circuit symbols.

A logic gate is a basic electronic circuit with one or more inputs and one output, which carries out a particular logical operation on binary signals. The output is high (1) or low (0) according to a fixed rule, described by its truth table. The basic gates are AND, OR and NOT, and NAND and NOR are their combinations.

OR gate. Its symbol is a curved 'shield' shape with two inputs, and its output is 1 when any input is 1; the output is 0 only when both inputs are 0. Boolean expression $Y = A + B$.

A	B	Y = A + B (OR)	Y = NAND (A·B)
0	0	0	1
0	1	1	1
1	0	1	1
1	1	1	0

Truth table of OR and NAND gates

NAND gate. It is an AND gate followed by a NOT gate, and its symbol is the AND symbol (a flat-backed 'D' shape) with a small circle at the output. Its output is 0 only when both inputs are 1, and 1 in all other cases. Boolean expression Y = (A·B) bar.
So the NAND gate is the exact opposite of the AND gate, and it is called a universal gate because AND, OR and NOT can all be built from NAND gates alone. Both gates are used to combine sensor signals in alarms, interlocks and control circuits.

19.5 How does a NOR gate work, and how can it be used to build other basic logic gates?

A NOR gate is an OR gate followed by a NOT gate, so it inverts the output of an OR gate. Its symbol is the OR symbol with a small circle at the output. Its output is 1 only when both inputs are 0; if any input is 1, the output is 0. Boolean expression Y = (A + B) bar.

A	B	Y = (A + B) bar (NOR)
0	0	1
0	1	0
1	0	0
1	1	0

Truth table of a two-input NOR gate

Using NOR gates to build the other gates:
1. NOT gate — join both inputs of a NOR gate together and feed the signal into it. Then Y = (A + A) bar = Ā, so the NOR gate inverts the input.
2. OR gate — take the output of a NOR gate and pass it through a NOR gate used as a NOT gate; this double inversion gives Y = A + B.
3. AND gate — apply De Morgan's theorem, A·B = (Ā + B̄) bar. Invert A and invert B with two NOR gates used as NOT gates, then feed both into a third NOR gate, and the result is A·B.
So NOR, like NAND, is a universal gate: any logic circuit can be built from NOR gates alone. This is why some logic families (such as the old RTL and CMOS families) were designed entirely around NOR gates — one type of gate is enough to make every decision circuit.

19.6 How can a burglar alarm be designed using a NAND gate and an LDR? Explain.

Components. A NAND gate with two inputs A and B, a light-dependent resistor (LDR), a push-button switch S, a battery, two resistors to form the input connections, and an alarm (buzzer) at the output.
Connections. The LDR is connected between input B of the NAND gate and the positive terminal of the battery, with a resistor from B to the negative line. The push-button switch S is connected between input A and the positive terminal, again with a resistor to the negative line. The alarm is connected at the output of the NAND gate.

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Working. In normal daylight the LDR has a low resistance, so input B is high (1). With the switch S open (not pressed) input A is also high (1). A NAND gate gives a 0 output only when both its inputs are 1, so in this safe condition the output is low and the alarm does not sound.

If a burglar blocks the light falling on the LDR, its resistance becomes high and input B falls to 0; if he steps on the switch S, input A falls to 0; if both happen, both inputs are 0. In every one of these cases at least one input is not 1, so the NAND output goes high (1) and the alarm rings. The circuit is therefore fail-safe: any interruption of the guarded condition sets off the alarm. The sensitivity is set with the two resistors, and the LDR is placed where the light beam can be cut.

Points to Remember

- Diode: conducts in forward bias (barrier ≈ 0.7 V for Si), blocks in reverse bias.
- LED: forward-bias recombination gives light; colour depends on the band gap of the material.
- Digital signal: only two levels, 0 and 1; 8 bits = 1 byte.
- Gates: AND (·), OR (+), NOT (bar); NAND and NOR are universal gates.
- NOR gives 1 only when all inputs are 0; NAND gives 0 only when all inputs are 1.

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